SAMSUNG

Chapter Quiz

Coding and Programming

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Quiz

Chapter 5

For Student

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Q. 27-01

How many swaps were executed in the bubble sort process below?

```python
def bubblesort(S):
    n = len(S)
    for i in range(n):
        print(S)
        for j in range(n - 1):
            if S[j] > S[j + 1]:
                S[j], S[j + 1] = S[j + 1], S[j]
```

```python
S = [50, 30, 40, 10, 20]
bubblesort(S)
print(S)
```

```
[50, 30, 40, 10, 20]
[30, 40, 10, 20, 50]
[30, 10, 20, 40, 50]
[10, 20, 30, 40, 50]
[10, 20, 30, 40, 50]
[10, 20, 30, 40, 50]
```

```
/home/eduardo/IdeaProjects/Curso-de-Python-EOI/
[50, 30, 40, 10, 20]
[30, 40, 10, 20, 50]
[30, 10, 20, 40, 50]
[10, 20, 30, 40, 50]
[10, 20, 30, 40, 50]
[10, 20, 30, 40, 50]
Total de swaps ejecutados: 8
```

Q. 27-02

How many comparisons were executed in the insertion sort process below?

```python
def insertionsort2(S):
    n = len(S)
    for i in range (1, n):
        print(S)
        x = S[i]
        j = i - 1
        while j >= 0 and S[j] > x:
            S[j + 1] = S[j]
            j -= 1
        S[j + 1] = x
```

```python
S = [50, 30, 40, 10, 20]
insertionsort2(S)
print(S)
```

```
[50, 30, 40, 10, 20]
[30, 50, 40, 10, 20]
[30, 40, 50, 10, 20]
[10, 30, 40, 50, 20]
[10, 20, 30, 40, 50]
```

```
/home/eduardo/IdeaProjects/Curso-de-Python-EOI/
[50, 30, 40, 10, 20]
[30, 50, 40, 10, 20]
[30, 40, 50, 10, 20]
[10, 30, 40, 50, 20]
[10, 20, 30, 40, 50]
Total de comparaciones ejecutadas: 10
```

Q. 28-01

How many times was the merge2() function executed in the merge sort process below?

```
S = [6, 2, 11, 7, 5, 4, 8, 16, 10, 3]
mergesort2(S, 0, len(S) - 1)
print(S)
```

```
/home/eduardo/IdeaProjects/Curso-de-Python-EOI/
La función merge2() se ejecutó 9 veces.

Process finished with exit code 0
```

Q. 29-01

Given the list below, write the output after executing the partition1() function.

```
S = [15, 10, 12, 20, 25, 13, 22]
partition1(S, 0, len(S) - 1)
print(S)
```

```
/home/eduardo/IdeaProjects/Curso-de-Python-EOI/Chapter03/.venv/bin/
Resultado después de partition1: [13, 10, 12, 15, 25, 20, 22]

Process finished with exit code 0
```